

Cross-Dataset Generalization of Hate-Speech Classifiers

Llama 3.2 3B Instruct and Mistral 7B Instruct v0.3

Final revised experimental report with completed cross-dataset evaluation

Abstract

This study tests whether supervised fine-tuning (SFT) on one hate-speech dataset produces knowledge that transfers to another dataset. Llama 3.2 3B Instruct and Mistral 7B Instruct v0.3 were evaluated before and after one epoch of LoRA-based 4-bit fine-tuning on HateXplain. In-domain performance improved for both models: fine-tuned Llama reached 56.91% accuracy and 49.67% macro F1, while fine-tuned Mistral reached 60.86% accuracy and 58.87% macro F1 under a deterministic relaxed parser. The completed external experiment evaluates the same baseline and fine-tuned models on a balanced 3,267-example sample from the Implicit Hate corpus, containing equal numbers of explicit hate, implicit hate, and non-hate posts. The transfer results are model-dependent. Llama's external macro F1 fell from 57.29% to 47.33% after SFT, while Mistral's rose from 29.55% to 47.28%. The clearest shared result is weak transfer to implicit hate: after SFT, implicit-hate recall was 11.20% for Llama and 7.44% for Mistral. These results show that strong in-domain improvement does not guarantee strong cross-dataset generalization, and that indirect or coded hate remains much harder to transfer than overt hate.

Keywords - hate speech, implicit hate, supervised fine-tuning, LoRA, cross-dataset evaluation, domain shift, Llama, Mistral.

1. Introduction and Research Questions

Hate-speech classification is not only a question of fitting one benchmark. Different datasets use different platforms, collection rules, target groups, and label definitions. A model can learn useful patterns on its training dataset and still fail when the style or definition changes. This is especially important for implicit hate, where the hateful meaning may depend on stereotypes, coded references, sarcasm, or context rather than a direct slur.

The original project focused on whether one epoch of parameter-efficient SFT improves classification. In the revised study, one epoch is treated only as a controlled and resource-aware training configuration. The main research question is broader: How does SFT on HateXplain affect in-domain classification, and how well does that learned behavior generalize to an independent dataset with explicit and implicit hate?

The analysis asks four secondary questions: (1) Does SFT improve the original HateXplain task? (2) Does that improvement transfer to another corpus? (3) Is transfer different for explicit and implicit hate? (4) Do Llama and Mistral show the same transfer pattern? These questions directly test whether the learned behavior is dataset-specific or more general.

2. Related Work and Current Dataset Status

HateXplain was introduced by Mathew et al. as an explainable hate-speech benchmark. Each post has a three-class label, target-community information, and human rationales. The dataset was designed to support both classification and analysis of why a model makes a decision [1]. Cross-dataset work later showed that hate-speech models often learn dataset-specific biases, and that performance can drop when training and test corpora differ [5]. This makes external evaluation an important part of a reliable hate-speech study.

The Latent Hatred work by ElSherief et al. focuses on a different difficulty: much previous research concentrated on overt hate, while real social-media hate can be indirect or coded. Their corpus and taxonomy were created to study implicit hate and the meanings behind it [3]. This provides a useful external test for a model trained only on HateXplain.

Current web status was checked on 20 August 2026. The Hugging Face HateXplain repository still lists the dataset as English text classification under CC BY 4.0 and reports the same official split sizes used in this project: 15,383 train, 1,922 validation, and 1,924 test examples [2]. The current tasksource/implicit-hate-stg1 repository exposes one 21,480-row train split with three labels and fields post and class [4]. Therefore, the dataset counts used in this report match both the completed experiment and the currently published dataset pages.

3. Datasets and Label Meaning

3.1 HateXplain

HateXplain contains English social-media posts collected from Twitter and Gab. The text is short and informal and may contain hashtags, mentions, spelling variation, slang, profanity, and identity terms. Each post was annotated by three crowd workers. Annotators provided a class label, identified the target community when relevant, and marked rationale tokens. The project reconstruction uses a two-of-three majority label and keeps the official train/validation/test divisions.

The three labels are meaningfully different. Hate describes identity-based hostility toward a person or group. Offensive describes rude, insulting, abusive, or profane language that does not satisfy the hate definition. Normal contains neither hate nor offensive content. This distinction is important because a system that simply treats all offensive language as hate will over-predict hate and ignore the dataset's intended taxonomy.

Class	Train	Validation	Test	Total
Hate	4,748	593	594	5,935
Normal	6,251	781	782	7,814
Offensive	4,384	548	548	5,480
All	15,383	1,922	1,924	19,229

Table 1. HateXplain counts after majority-label reconstruction and official splitting.

Dataset class distributions

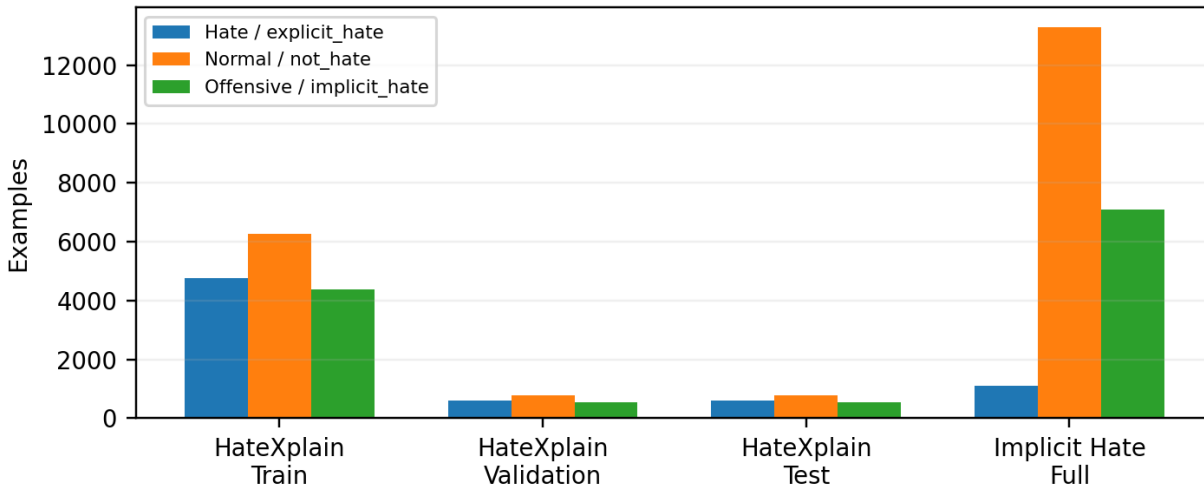


Figure 1. Dataset class distributions. For the external corpus, the bars show the current full stage-1 source labels.

3.2 Implicit Hate / Latent Hatred

The external dataset is tasksource/implicit-hate-stg1, a text-classification version of the Latent Hatred corpus. As currently published, it contains 21,480 English posts in one public train split: 1,089 explicit_hate, 7,100 implicit_hate, and 13,291 not_hate. This natural distribution is strongly imbalanced, with explicit hate forming only 5.07% of the corpus.

Explicit hate communicates hostility directly. Implicit hate communicates prejudice indirectly, for example through stereotypes, coded language, grievance, inferiority claims, irony, threats, or incitement. Detecting it can require more semantic and contextual reasoning than recognizing an overt attack. The external experiment therefore gives a stronger test of generalization than a second random split from the same HateXplain distribution.

3.3 Balanced external evaluation and binary mapping

The full 21,480 examples are used for descriptive statistics, but external inference uses a deterministic balanced sample with seed 42. The sample takes all 1,089 explicit-hate examples and exactly 1,089 examples from each of the other two source classes, giving 3,267 examples total. The same cached sample IDs are used for all four conditions. Balancing is useful here because the goal is to compare behavior across the three source classes rather than to estimate deployment accuracy under the natural prevalence.

The two datasets do not share the same label taxonomy, so a common binary layer is required for the cross-dataset test. HateXplain Hate maps to HATE. HateXplain Normal and Offensive map to NON-HATE, because offensive language is not automatically identity-based hate. In the external corpus, explicit_hate and implicit_hate map to HATE, while not_hate maps to NON-HATE. The original external source label is preserved so recall can still be reported separately for explicit hate, implicit hate, and not-hate.

Source label	Full count	Full share	Evaluation count	Binary target
explicit_hate	1,089	5.07%	1,089	HATE
implicit_hate	7,100	33.05%	1,089	HATE
not_hate	13,291	61.88%	1,089	NON-HATE
All	21,480	100%	3,267	-

Table 2. Current Implicit Hate stage-1 distribution and the balanced evaluation sample used in this study.

4. Models and Method

4.1 Models and fine-tuning

The evaluated repositories are unsloth/Llama-3.2-3B-Instruct-bnb-4bit and unsloth/mistral-7b-instruct-v0.3-bnb-4bit. The project uses 4-bit quantized instruction-tuned causal language models and LoRA adapters. LoRA updates a small set of low-rank adapter parameters while the base model weights remain frozen, which reduces memory and training cost [6]. The trained adapter size was about 24.3 million parameters for Llama and 41.9 million for Mistral.

Both models were trained for one SFT epoch on the same HateXplain training data with per-device batch size 2 and gradient accumulation 4, giving an effective batch size of 8. The final notebooks use completion-only token masking so optimization is focused on the assistant classification response. One epoch was chosen to keep the comparison

feasible and controlled; the study does not claim that one epoch is optimal.

The underlying model families remain available in current repositories. The current Unsloth Llama 3.2 3B Instruct 4-bit repository identifies the model as conversational text generation with bitsandbytes 4-bit quantization [9]. The official Mistral 7B Instruct v0.3 repository currently describes a 7B instruct model with a 32,768-token vocabulary, v3 tokenizer support, and function-calling support [10]. These current repository descriptions are used only for model documentation; the reported scores come from the archived project checkpoints and predictions.

4.2 Classification prompt and parsing

The classification instruction asks the model to return exactly one of three labels: Hate, Normal, or Offensive, and to return only the label. The same semantic prompt is used for both model families, while each tokenizer uses its own chat

template. This keeps the task definition stable while respecting model-specific conversation formatting.

The first evaluation used strict output matching. Post-analysis showed that fine-tuned Mistral frequently repeated the same valid class label. A deterministic relaxed parser was therefore used for the corrected Mistral in-domain result and for both models in the external experiment: one valid label is accepted; repeated copies of the same label are collapsed; two different valid labels are invalid; and outputs with no valid label are invalid. This rule does not guess between contradictory labels. Invalid outputs count as incorrect in end-to-end accuracy.

4.3 Evaluation metrics

For HateXplain, the main metrics are accuracy, macro precision, macro recall, macro F1, per-class F1, and invalid-output rate. For cross-dataset evaluation, the main metrics are binary accuracy, binary macro precision/recall/F1, HATE

precision/recall/F1, invalid-output rate, and source-class recall for explicit hate, implicit hate, and not-hate. Macro F1 is important because it gives equal weight to both binary classes even when the evaluation composition is not naturally balanced after mapping.

5. In-Domain Results on HateXplain

Fine-tuning improved the aggregate HateXplain results for both model families. Llama accuracy increased from 34.41% to 56.91% and macro F1 from 39.19% to 49.67%. Fine-tuned Mistral achieved the strongest aggregate in-domain scores after the relaxed-parser correction: 60.86% accuracy and 58.87% macro F1. The original strict Mistral result is not discarded; it is treated as an output-format diagnostic because most rejected responses were repeated labels rather than unrelated text.

Model	Stage	Accuracy	Macro F1	Invalid
Llama 3.2 3B	Baseline	34.41%	39.19%	24.53%
Llama 3.2 3B	SFT	56.91%	49.67%	2.91%
Mistral 7B	Baseline	35.86%	29.21%	2.13%
Mistral 7B	SFT (relaxed)	60.86%	58.87%	11.85%

Table 3. Verified HateXplain test results, $N = 1,924$ per condition.

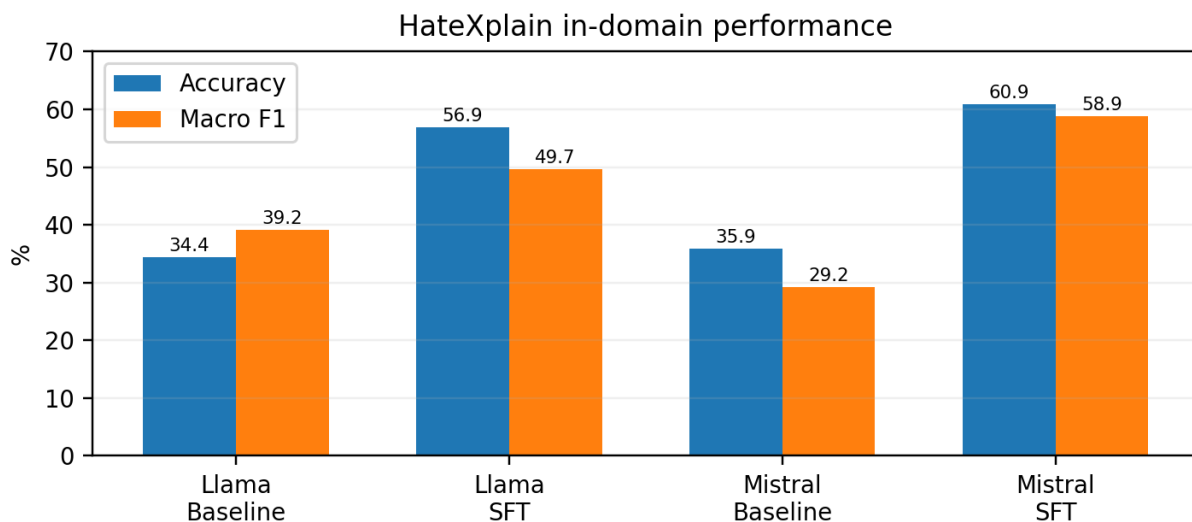


Figure 2. In-domain accuracy and macro F1. Fine-tuned Mistral uses the deterministic relaxed parser.

Class-level analysis shows why aggregate scores are not enough. Fine-tuned Llama reached 66.29% F1 for Hate and 66.63% for Normal, but only 16.08% for Offensive. Fine-tuned Mistral reached 73.81% F1 for Hate, 72.09% for Normal, and 30.70% for Offensive. In both models, Offensive remains the weakest class after adaptation. This supports the lecturer's concern that the report should focus on language behavior and class distinctions, not mainly on formatting failures.

6. Completed Cross-Dataset Results

The external experiment was completed for all four conditions on the same 3,267-example balanced sample. No model was trained on the Implicit Hate corpus. Therefore, the comparison isolates transfer from HateXplain rather than adaptation to the external data.

Condition	Accuracy	Macro F1	Explicit R	Implicit R	Not-hate R	Invalid
Llama baseline	53.87%	57.29%	57.85%	47.57%	56.20%	15.82%
Llama SFT	47.72%	47.33%	38.02%	11.20%	93.94%	3.37%
Mistral baseline	34.89%	29.55%	7.25%	3.76%	93.66%	0.00%
Mistral SFT	44.69%	47.28%	34.25%	7.44%	92.38%	16.80%

Table 4. Completed cross-dataset results on the balanced Implicit Hate sample, $N = 3,267$ per condition.

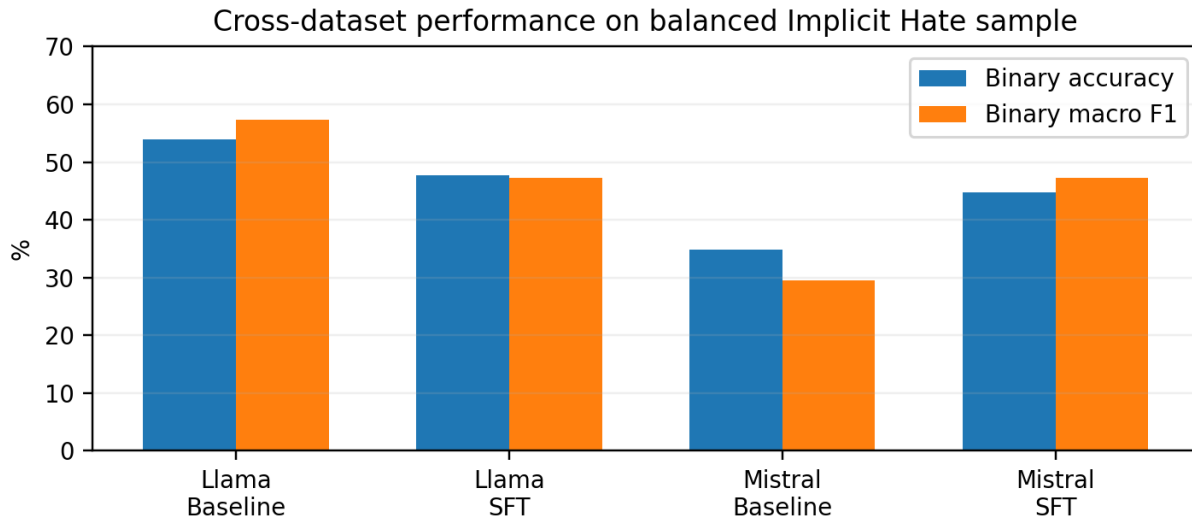


Figure 3. Cross-dataset binary accuracy and macro F1.

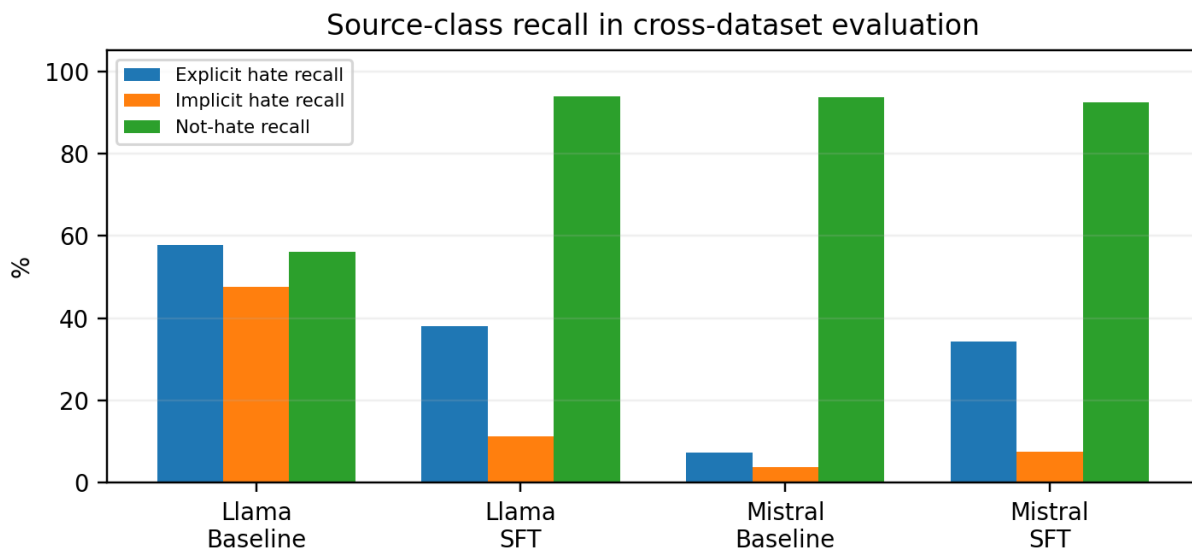


Figure 4. Recall by original external source class.

6.1 Llama transfer behavior

Llama shows the clearest example of in-domain improvement without external improvement. After HateXplain SFT, external accuracy fell from 53.87% to 47.72%, and macro F1 fell from 57.29% to 47.33%. The main change is a strong shift toward NON-HATE. Not-hate recall rose from 56.20% to 93.94%, but explicit-hate recall fell from 57.85% to 38.02% and implicit-hate recall fell sharply from 47.57% to 11.20%. Hate precision rose to 97.28%, but hate recall fell to 24.61%. This means the model became much more conservative: when it predicts hate it is usually correct, but it misses many hateful examples.

This is important because the in-domain HateXplain result alone would suggest that the adaptation was broadly successful. The external test shows that the improvement was not fully transferable. One plausible interpretation is that the model learned decision boundaries that fit HateXplain's wording and three-class taxonomy, especially the separation between Normal and Offensive, but those boundaries do not cover the different cues used in the implicit-hate corpus.

6.2 Mistral transfer behavior

Mistral shows a different pattern. Its baseline external performance is weak: 34.89% accuracy, 29.55% macro F1, 7.25% explicit-hate recall, and 3.76% implicit-hate recall. After

SFT, accuracy rises to 44.69% and macro F1 to 47.28%.

Explicit-hate recall increases substantially to 34.25%, while implicit-hate recall only reaches 7.44%. Not-hate recall stays very high, changing only from 93.66% to 92.38%.

Therefore, Mistral gains meaningful transfer for overt hate but almost no transfer for implicit hate. Its hate precision after SFT is very high at 97.22%, while hate recall is only 20.84%. The fine-tuned model also has a 16.80% invalid-output rate on the external sample. This again separates two issues: the semantic pattern is more useful than baseline, but free-form generation still creates an interface-compliance cost.

7. Discussion

7.1 Main finding: in-domain success does not guarantee transfer

The strongest conclusion is that a better score on the training-domain benchmark does not guarantee better performance on a different hate-speech corpus. This matches the broader cross-dataset literature, which reports that dataset construction and annotation choices can create dataset-specific biases [5]. Llama demonstrates the issue directly: its HateXplain accuracy improves by more than 22 percentage points after SFT, but its external macro F1 decreases by about

10 points. Mistral improves in both domains, but the size and type of improvement are different.

7.2 Implicit hate is the hardest transfer case

Both model families remain poor at implicit hate after HateXplain fine-tuning. Llama reaches only 11.20% implicit-hate recall after SFT, and Mistral only 7.44%. These values are far below their not-hate recall and below Llama's baseline implicit-hate recall. This supports the motivation of the Latent Hatred benchmark: indirect hate is not simply a weaker version of explicit hate. It can require understanding stereotypes, implications, coded references, and social context [3]. Training on a dataset where the main classes are Hate, Offensive, and Normal does not automatically teach those distinctions.

The result is also consistent with later research that treats cross-domain robustness as a separate problem rather than assuming it follows from single-dataset accuracy. Antypas and Camacho-Collados found that different hate-speech datasets have different generalization properties and that combining datasets can improve robustness [5]. Our smaller experiment reaches a compatible qualitative conclusion: the source dataset matters, and transfer should be measured directly.

7.3 Llama and Mistral learn different transfer patterns

The two models do not respond to the same SFT procedure in the same way. Llama becomes much better at rejecting non-hate, but at the cost of missing hateful external examples. Mistral starts from a much weaker external hate detector and gains explicit-hate sensitivity after SFT. This model-specific behavior is itself important: there is no single statement that 'SFT improves generalization.' The answer depends on the model and on which type of hate is measured.

7.4 Output validity is secondary, not the main research question

The earlier report gave too much attention to invalid outputs. The completed study treats them as a methodological measurement rather than the main evidence about language understanding. The relaxed parser solves one specific formatting artifact - repeated identical labels - without accepting contradictory outputs. Better engineering could further reduce this problem through constrained decoding, stop tokens, structured schemas, or a classifier head. These changes may improve interface reliability, but they would not by themselves solve the much larger cross-dataset gap on implicit hate.

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8. Limitations and Future Work

The study uses one SFT epoch, one hyperparameter configuration, and one training seed for each model. More epochs or a hyperparameter search could change the result, so the report does not claim that the selected training configuration is optimal. The external evaluation uses a balanced sample rather than the natural class prevalence. This is useful for class comparison but means the reported external accuracy should not be read as an estimate of natural-corpus deployment accuracy.

The label mapping is necessary but imperfect. HateXplain Offensive is mapped to NON-HATE because offense is not necessarily identity-based hate, while the external corpus uses a binary hate/non-hate abstraction after preserving its original source class. Different mappings could change the binary scores. The two corpora also differ in collection process, platform mix, annotation policy, and time period, so the external performance change is a combined domain shift rather than a pure test of one linguistic factor.

Future work should test multi-dataset training or staged adaptation, because prior cross-dataset research suggests that combining corpora can improve robustness [5]. Another useful experiment would directly train on implicit-hate examples and test whether explicit-hate gains can be preserved. Structured decoding should also be tested separately so formatting reliability is not confused with semantic classification.

9. Conclusion

The revised experiment answers a stronger question than the original one-epoch framing. Fine-tuning on HateXplain improves in-domain aggregate performance for both Llama and Mistral, but the new external evaluation shows that this improvement is not automatically transferable. Llama becomes more conservative and loses substantial external hate recall, while Mistral improves external macro F1 and explicit-hate recall. Most importantly, both fine-tuned models remain very weak on implicit hate. The main lesson is therefore not simply that SFT helps or hurts. It is that hate-speech performance is strongly affected by dataset definitions and domain shift, and that implicit hate should be evaluated separately because overt-hate improvements do not reliably transfer to coded or indirect hate.

Appendix A. Lecturer Feedback Coverage

Lecturer concern	How the revised report addresses it
Why one epoch?	One epoch is now a controlled resource-aware setting, not the main research question.
Dataset description too slim	Sections 3.1-3.3 explain source, genre, annotation, label meaning, targets, counts, and current published status.
Invalid outputs over-emphasized	Validity is separated from semantic performance and moved to a methodological role.
Prompt not specified	Section 4.2 explains the exact three-label format and parsing rules.
Need more than one dataset	Completed cross-dataset evaluation uses the Implicit Hate corpus.
Explicit vs implicit hate	Table 4 and Figure 4 report source-class recall separately.
Does learning transfer?	Sections 6-7 directly compare baseline and HateXplain-SFT models on external data.
Reference points from prior work	Section 2 and Section 7 connect the findings to HateXplain, Latent Hatred, and cross-dataset robustness literature.

Table A1. Traceability from lecturer feedback to the revised experimental report.

Appendix B. Verified Cross-Dataset Metric Detail

Condition	HATE precision	HATE recall	HATE F1	Invalid count
Llama baseline	78.52%	52.71%	63.08%	517
Llama SFT	97.28%	24.61%	39.28%	110
Mistral baseline	63.49%	5.51%	10.14%	0
Mistral SFT	97.22%	20.84%	34.33%	549

Table B1. Additional verified metrics recomputed from the uploaded cross-dataset summaries.

The four external conditions used exactly 3,267 evaluation examples each. The uploaded result archives also include prediction-level CSV files, invalid-output subsets, error subsets, confusion-matrix images, metric JSON files, dataset statistics, and run manifests. The balanced sample contains 1,089 examples from each source class. These artifacts make it possible to audit the headline values without rerunning GPU inference.

Appendix C. Reproducibility Checklist

- Use seed 42 and the cached balanced Implicit Hate sample with 1,089 examples per source class.
- Use the same cached sample IDs for Llama baseline, Llama SFT, Mistral baseline, and Mistral SFT.
- Preserve raw generated text before parsing and keep INVALID as an evaluation outcome, not a semantic class.
- Apply the same deterministic relaxed-parser rule in the external evaluation for both model families.
- Reuse the completed HateXplain adapters; the cross-dataset experiment performs inference only and does not retrain them.
- Report natural full-corpus statistics separately from the balanced evaluation sample.